IEOR 221 – Homework 4 Student ID: 3040728564

Module 5: Options – Basics

1. Arbitrage with a European Put (Hull 11.17)

$$S_0 = 47, \quad K = 50, \quad r = 0.06, \quad T = \tfrac{1}{12}, \quad p_0 = 2.50$$

For a non-dividend stock, put–call parity gives

$$c_0 + Ke^{-rT} = p_0 + S_0.$$

Compute

$$Ke^{-rT} = 50e^{-0.06/12} = 50 \times 0.9950 = 49.75, \qquad c_0 = 2.50 + 47 - 49.75 = -0.25.$$

Because $c_0 < 0$ is impossible, the put is overpriced.

Arbitrage: short one put (+2.50), buy the stock (−47), and borrow $Ke^{-rT} = 49.75$. Net cash today:

$$2.50 - 47 + 49.75 = 5.25 > 0.$$

At T:

$$\begin{cases} S_T > 50: & \text{put expires worthless, repay 50, keep } S_T - 50 > 0, \\ S_T \le 50: & \text{put exercised, sell stock at 50, repay 50} \Rightarrow 0. \end{cases}$$

Hence a risk-free gain $\approx 5.25e^{0.06/12} = 5.27$.

2. European Put via Parity with Dividends (Hull 11.19)

$$S_0 = 29, \quad K = 30, \quad r = 0.10, \quad T = 0.5, \quad c_0 = 2.$$

Dividends: 0.5 at months 2 and 5.

$$PV(D) = 0.5e^{-0.10\times\frac{2}{12}} + 0.5e^{-0.10\times\frac{5}{12}} = 0.5(0.9835 + 0.9595) = 0.9715.$$

$$Ke^{-rT} = 30e^{-0.10\times0.5} = 30e^{-0.05} = 28.54.$$

Parity with dividends:

$$p_0 = c_0 + Ke^{-rT} - S_0 + PV(D) = 2 + 28.54 - 29 + 0.97 = 2.51.$$

3. Convexity of Call Prices (Hull 11.31)

Let $K_3 > K_2 > K_1$ with $K_3 - K_2 = K_2 - K_1 = h$. Form the portfolio

$$V(S_T) = (S_T - K_1)^+ + (S_T - K_3)^+ - 2(S_T - K_2)^+.$$

Piecewise evaluation:

$$\begin{aligned} &S_T \le K_1: && V = 0, \\ &K_1 < S_T \le K_2: && V = S_T - K_1 \ge 0, \\ &K_2 < S_T \le K_3: && V = (S_T - K_1) - 2(S_T - K_2) = -S_T + 2K_2 - K_1 \ge 0, \\ &S_T > K_3: && V = (S_T - K_1) + (S_T - K_3) - 2(S_T - K_2) = 0. \end{aligned}$$

Thus $V(S_T) \ge 0$ for all S_T. No-arbitrage $\Rightarrow c_1 + c_3 - 2c_2 \ge 0 \Rightarrow c_2 \le (c_1 + c_3)/2$.

4. Bear Spread with Puts (Hull 12.22)

Sell put $K_1 = 25$ for 2.15; buy put $K_2 = 29$ for 4.75.

$$\text{Initial outlay } = 4.75 - 2.15 = 2.60.$$

Payoff:

$$\Pi(S_T) = \max(K_2 - S_T, 0) - \max(K_1 - S_T, 0).$$

Hence

$$\Pi(S_T) = \begin{cases} 4, & S_T \leq 25, \\ 29 - S_T, & 25 < S_T < 29, \\ 0, & S_T \geq 29. \end{cases}$$

At given prices:

$$S_T = 23 \Rightarrow \Pi = 4, \quad S_T = 28 \Rightarrow \Pi = 1, \quad S_T = 33 \Rightarrow \Pi = 0.$$

5. Diagonal Spread Profit at T_1 (Hull 12.25)

Long $C(K_2, T_2)$ and short $C(K_1, T_1)$ with $T_2 > T_1$. At T_1 the short call expires; the long call still has time value.

$$\text{Profit}(S_{T_1}) = C(S_{T_1}, K_2, T_2 - T_1) - \max(S_{T_1} - K_1, 0) - \text{net premium grown}.$$

- If $K_2 > K_1$: position is **bearish**; profit decreases as S_{T_1} rises, limited loss for high prices.
- If $K_2 < K_1$: position is **bullish**; profit increases with S_{T_1}, limited downside for low prices.

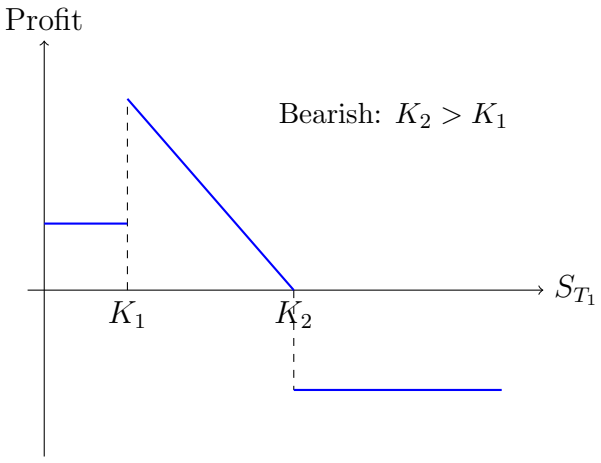


Figure 1: *
Bearish diagonal spread ($K_2 > K_1$): profit falls with S_{T_1}; capped loss beyond K_2.

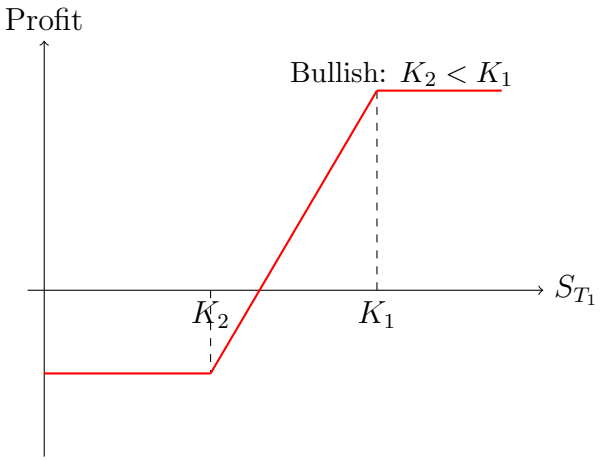


Figure 2: *
Bullish diagonal spread ($K_2 < K_1$): profit rises with S_{T_1}; limited loss for low prices.